

DIWALI SALES DATA ANALYSIS

Python-Based Exploratory Data Analysis Project

Project Report

Tools Used

Python • Pandas • NumPy • Matplotlib • Seaborn

Dataset: Diwali Sales Data

Records analyzed after cleaning: 11,239

Prepared from the supplied analysis notebook and dataset

1. Executive Summary

This project analyzes Diwali sales transaction data using Python to understand customer characteristics, purchasing behavior, geographical performance, occupations, and product categories. The analysis follows a practical data-analytics workflow: importing the dataset, inspecting its structure, cleaning missing and irrelevant data, converting data types, performing exploratory analysis, and visualizing the resulting patterns.

The original dataset contained 11,251 rows and 15 columns. Two completely blank/unrelated columns (Status and unnamed1) were removed, and 12 missing values in Amount were removed through row-wise deletion. The final analysis dataset contains 11,239 rows and 13 columns.

The supplied notebook identifies female customers, the 26–35 age group, major states such as Uttar Pradesh, Maharashtra and Karnataka, and occupations such as IT Sector, Healthcare and Aviation as important segments. Food, Clothing & Apparel, and Electronics & Gadgets are among the leading product categories by sales.

2. Project Objectives

- Understand the structure and quality of the Diwali sales dataset.
- Clean irrelevant columns and handle missing values.
- Analyze customer demographics such as gender, age group and marital status.
- Identify high-performing states and occupations.
- Evaluate product-category and product-level sales performance.
- Use visualizations to communicate major sales patterns.
- Derive actionable customer and product insights from the analysis.

3. Dataset Description

The dataset contains customer-level Diwali purchase information. The supplied notebook shows the following fields before cleaning:

Column	Description
User_ID	Customer identifier
Cust_name	Customer name
Product_ID	Product identifier
Gender	Customer gender
Age Group	Customer age band
Age	Customer age
Marital_Status	Encoded marital-status field
State	Customer state
Zone	Geographical zone
Occupation	Customer occupation
Product_Category	Purchased product category
Orders	Number of orders
Amount	Purchase amount

Initial dimensions: 11,251 rows × 15 columns. After cleaning: 11,239 rows × 13 columns.

4. Tools and Technologies

- Python – core programming and analysis environment.
- NumPy – numerical computing support.
- Pandas – data loading, cleaning, grouping and aggregation.
- Matplotlib – chart generation and plotting.
- Seaborn – statistical and categorical visualizations.
- Jupyter Notebook – interactive execution and documentation of the analysis.

5. Methodology

Data Import: The CSV file was loaded with Pandas using Unicode-compatible encoding.

Data Inspection: Shape, head, column information, data types and descriptive statistics were examined.

Column Cleaning: Status and unnamed1 were removed because they contained no usable values.

Missing-Value Handling: Amount contained 12 missing values; rows containing missing values were removed.

Data-Type Conversion: Amount was converted from floating-point to integer.

Descriptive Analysis: Age, Orders and Amount were summarized using describe().

Exploratory Data Analysis: Customer and sales patterns were examined by gender, age, state, marital status, occupation and product category.

Visualization: Bar charts and count plots were created using Matplotlib and Seaborn.

Conclusion: The strongest customer and product segments were identified from the observed patterns.

6. Data Cleaning and Preparation

The notebook first checked the dataset structure and found 15 columns. Status and unnamed1 were completely empty and were therefore removed. The Amount field had 12 missing values, which were handled by dropping the corresponding rows. Amount was then converted to integer format for analysis.

Stage	Rows	Columns
Original dataset	11,251	15
After removing blank columns	11,251	13
After removing missing Amount rows	11,239	13

7. Descriptive Statistics

Statistic	Age	Orders	Amount
count	11,239.00	11,239.00	11,239.00
mean	35.41	2.49	9,453.61
std	12.75	1.11	5,222.36
min	12.00	1.00	188.00
25%	27.00	2.00	5,443.00

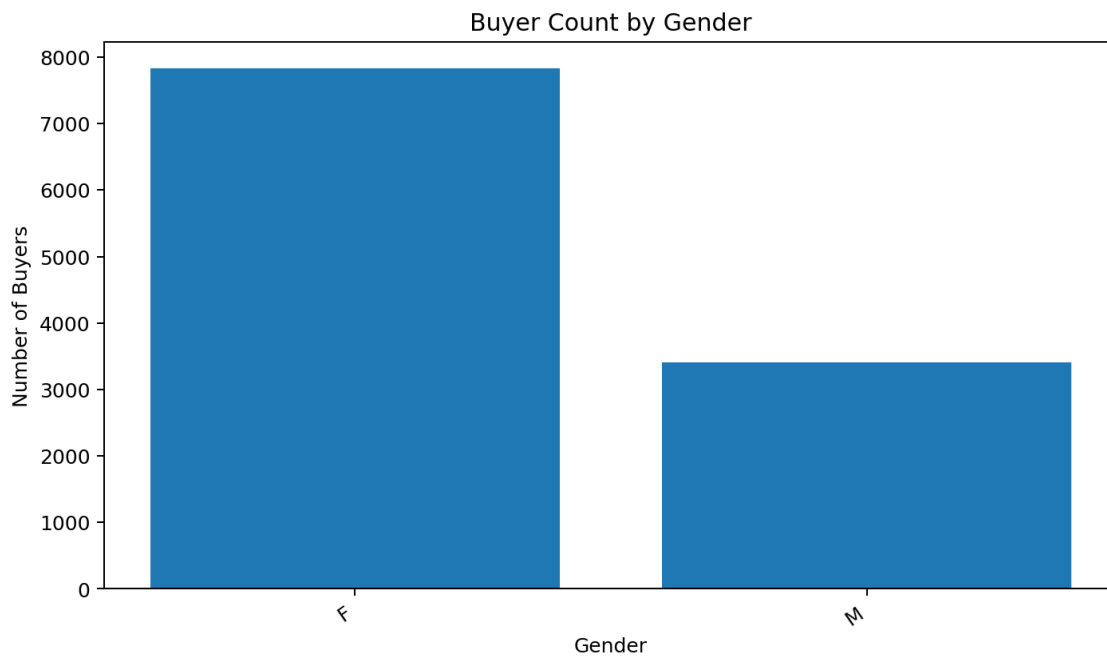
50%	33.00	2.00	8,109.00
75%	43.00	3.00	12,675.00
max	92.00	4.00	23,952.00

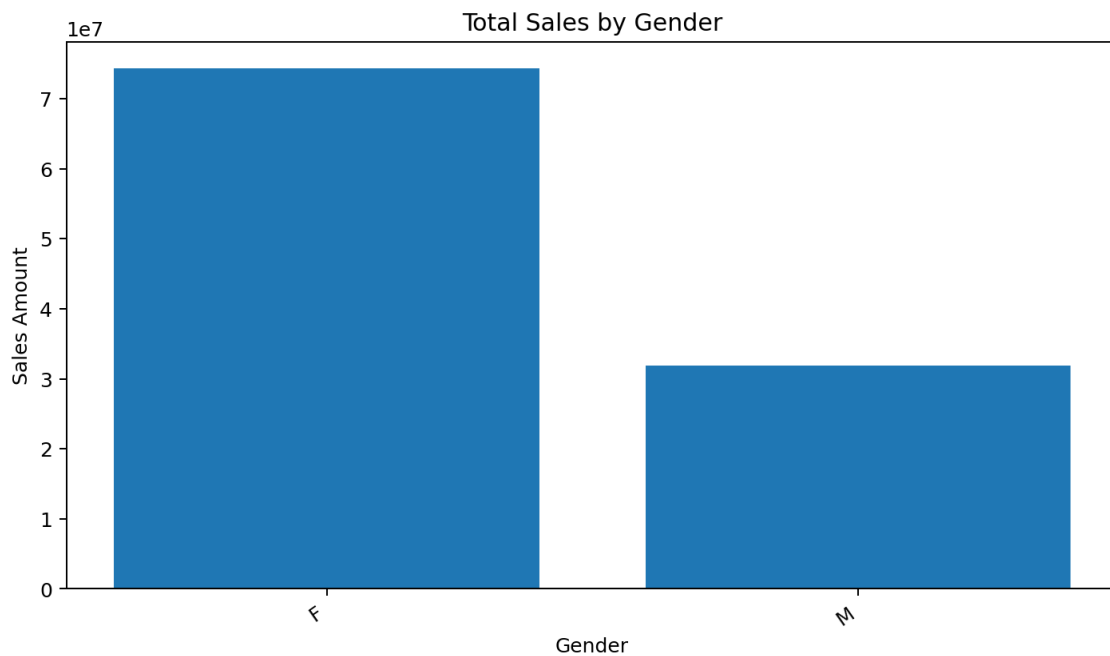
The mean customer age is 35.41 years, the average number of orders is 2.49, and the average transaction amount is ₹9,453.61.

8. Exploratory Data Analysis

8.1 Gender Analysis

The cleaned dataset contains 7,832 female records and 3,407 male records. Female customers account for ₹74,335,853 in sales compared with ₹31,913,276 for male customers.

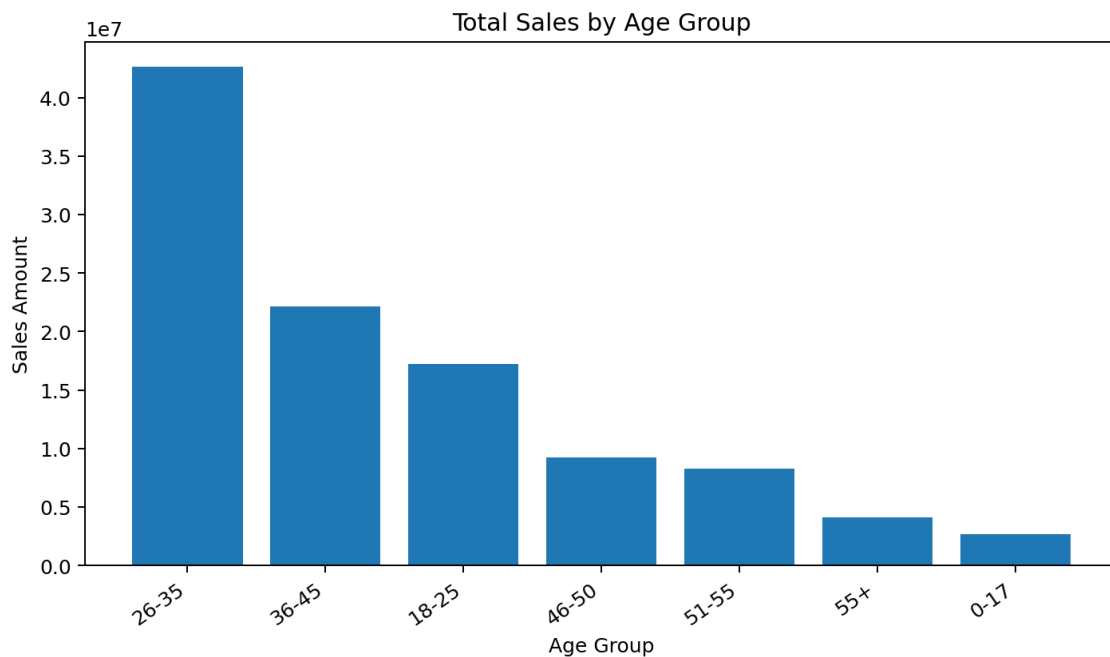




Notebook interpretation: most buyers are female, and the female segment also shows greater purchasing power.

8.2 Age Group Analysis

The 26–35 age group is the strongest sales segment, generating ₹42,613,442. It is followed by the 36–45 and 18–25 groups.

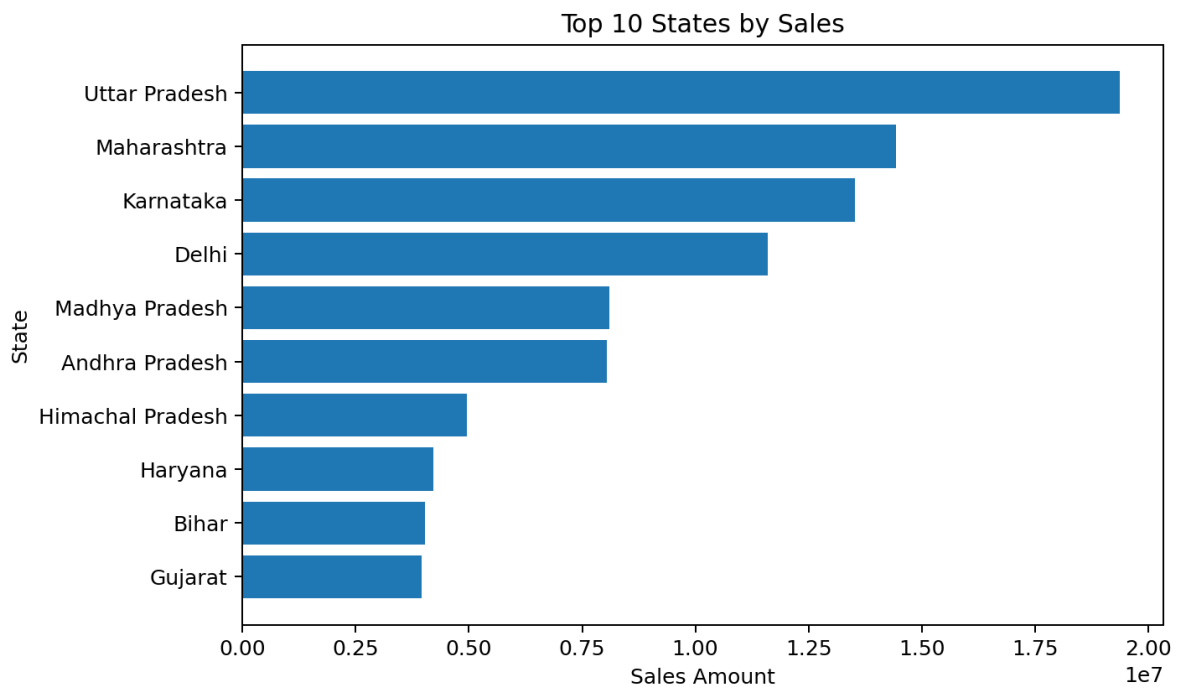


Notebook interpretation: buyers in the 26–35 age group, particularly females, form the dominant customer segment.

8.3 State Analysis

Geographically, Uttar Pradesh, Maharashtra and Karnataka are the three highest-sales states in the cleaned dataset.

Rank	State	Sales Amount
1	Uttar Pradesh	₹19,374,968
2	Maharashtra	₹14,427,543
3	Karnataka	₹13,523,540
4	Delhi	₹11,603,818
5	Madhya Pradesh	₹8,101,142



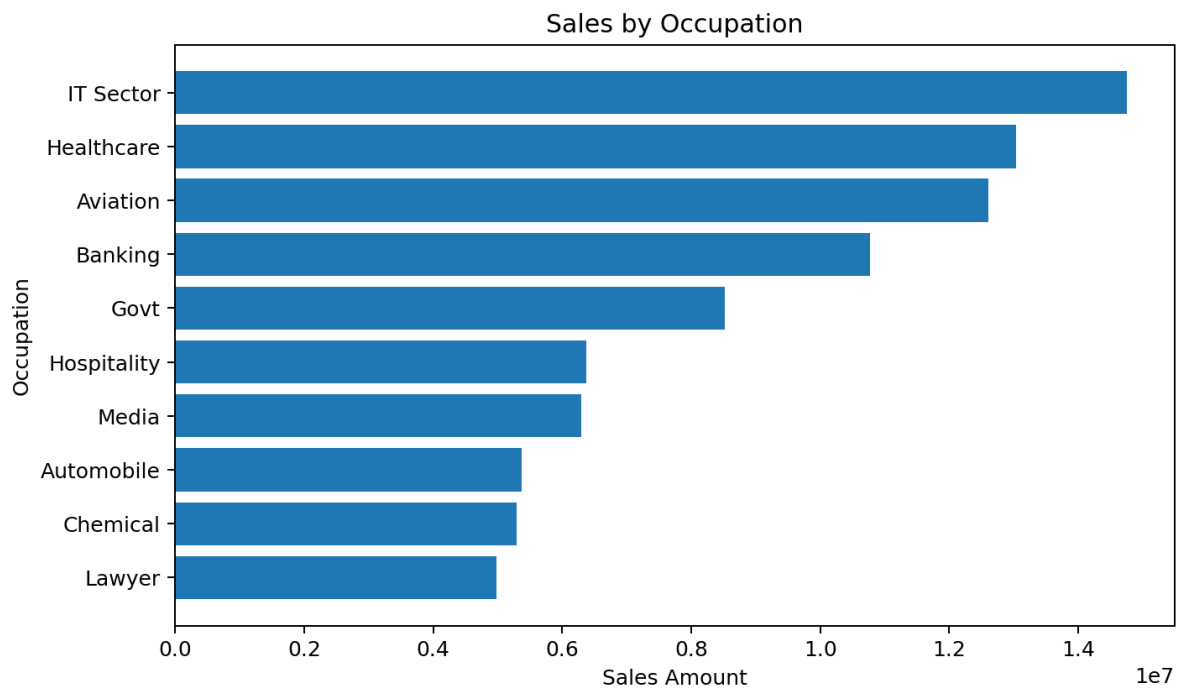
8.4 Marital Status Analysis

The notebook compares customer counts and total sales by Marital_Status, including a gender split. The source data stores marital status as numeric codes (0 and 1), and the supplied notebook uses this field to identify the dominant customer segment. Because the dataset itself does not provide a textual mapping for these codes, this report retains the encoded values rather than assigning an unsupported label.

- Marital_Status = 0: ₹62,125,384 total sales.
- Marital_Status = 1: ₹44,123,745 total sales.

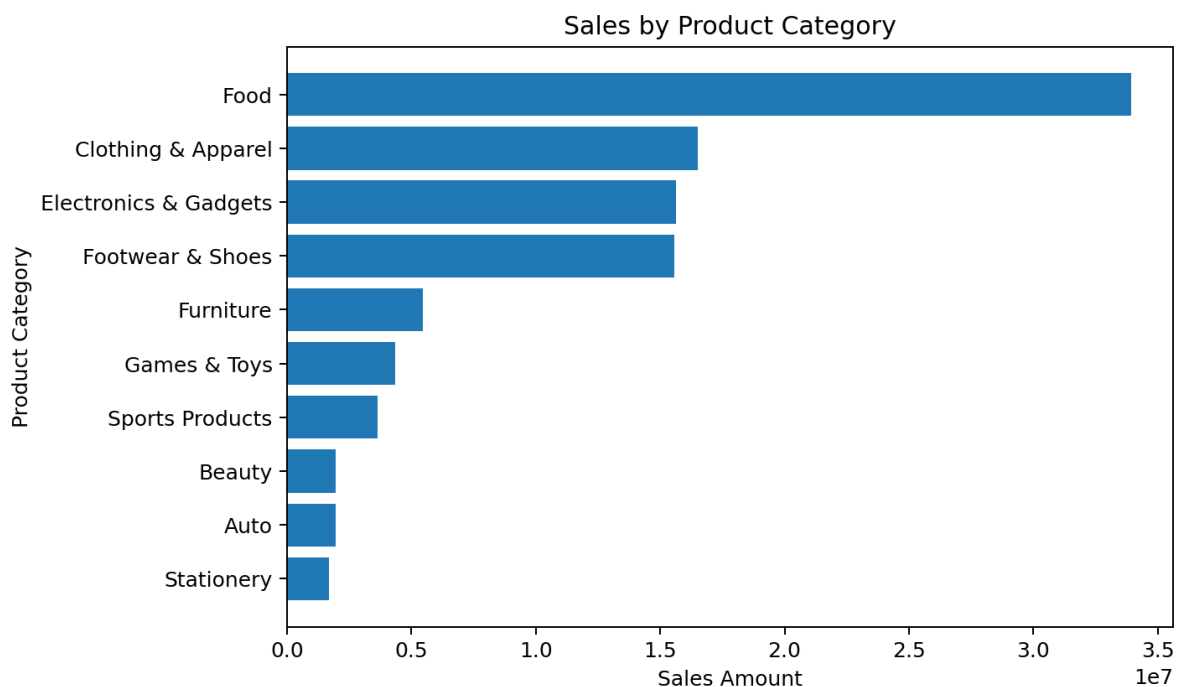
8.5 Occupation Analysis

The highest-sales occupations are IT Sector, Healthcare, and Aviation. The notebook similarly highlights IT Sector, Healthcare and Aviation as important buyer segments.



8.6 Product Category Analysis

Food is the highest-sales product category at ₹33,933,883, followed by Clothing & Apparel at ₹16,495,019 and Electronics & Gadgets at ₹15,643,846.



Notebook interpretation: Food, Clothing & Apparel, and Electronics & Gadgets are among the most sold categories.

8.7 Product-Level Analysis

Rank	Product ID	Orders	Sales
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1	P00265242	127	₹540,136
2	P00110942	116	₹424,833
3	P00237542	91	₹322,363
4	P00184942	82	₹401,816
5	P00025442	79	₹279,488
6	P00114942	79	₹254,641
7	P00145042	76	₹275,099
8	P00117942	76	₹281,013
9	P00044442	75	₹273,954
10	P00110842	74	₹290,661

9. Key Findings

- The cleaned dataset contains 11,239 usable records.
- Female customers form the larger buyer segment and contribute substantially more total sales than male customers.
- The 26–35 age group is the leading segment, with ₹42,613,442 in sales.
- Uttar Pradesh, Maharashtra and Karnataka are the top three states by total sales.
- IT Sector, Healthcare and Aviation are among the strongest occupation segments.
- Food, Clothing & Apparel and Electronics & Gadgets are the leading product categories by sales.
- The analysis demonstrates the usefulness of grouping and aggregation for identifying customer and product segments.

10. Business Recommendations

- Prioritize marketing campaigns toward the strongest demographic segments identified in the analysis.
- Develop targeted offers for customers aged 26–35, while maintaining differentiated campaigns for other age groups.
- Increase regional promotional focus in Uttar Pradesh, Maharashtra and Karnataka because these states generate the highest sales.
- Create occupation-oriented campaigns for high-performing professional groups such as IT Sector, Healthcare and Aviation.
- Maintain strong inventory and promotional visibility for Food, Clothing & Apparel, and Electronics & Gadgets.
- Use product-level order rankings to identify products suitable for bundles, cross-selling and promotional placement.
- Repeat the analysis periodically with additional time, discount, profit and customer-retention fields if available.

11. Limitations

- The dataset does not provide a time/date field in the analyzed columns, so sales trends over dates or festival days cannot be evaluated.
- Marital_Status is encoded numerically and the supplied files do not define the code mapping.
- The notebook uses row deletion for the 12 missing Amount values; this is appropriate for the existing workflow but may reduce available observations.

- The analysis focuses on sales and orders; profit, discount, cost and customer lifetime value are not available for evaluation.
- The results describe patterns in the supplied dataset and should not automatically be generalized beyond it.

12. Conclusion

The Diwali Sales Data Analysis project demonstrates a complete exploratory data-analysis workflow using Python. The project begins with raw transactional data, performs practical cleaning and type conversion, and then applies aggregation and visualization to understand customer and product behavior.

The supplied notebook's overall conclusion is that a strong customer segment is associated with women in the 26–35 age range, customers from Uttar Pradesh, Maharashtra and Karnataka, and occupations such as IT, Healthcare and Aviation, with Food, Clothing & Apparel and Electronics & Gadgets among the important product categories. These findings can support targeted marketing, regional planning and product-focused promotional strategies.

13. Appendix – Analysis Workflow

1. Import NumPy, Pandas, Matplotlib and Seaborn.
2. Read the Diwali sales CSV file.
3. Inspect dimensions and sample records.
4. Inspect data types and missing values.
5. Drop Status and unnamed1.
6. Drop rows with missing Amount.
7. Convert Amount to integer.
8. Generate descriptive statistics.
9. Analyze Gender and Age Group.
10. Analyze State and Marital_Status.
11. Analyze Occupation and Product_Category.
12. Identify top Product_ID values by orders.
13. Summarize findings and conclude.

End of Project Report